

AHLHORN, Germany

WMO: 102180

Lat: 52.883N

Lon: 8.233E

Elev: 56

StdP: 100.65

Time Zone: 1.00 (EUC)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.9	-8.1	-12.9	1.2	-10.1	-10.5	1.5	-6.9	14.6	8.3	13.3	7.2	3.0	70	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	29.5	19.3	27.2	18.4	25.2	17.4	20.5	27.3	19.3	25.3	18.3	23.7	3.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.0	23.0	17.0	12.2	21.9	16.1	11.5	20.9	58.8	27.0	55.0	24.9	51.8	23.7	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.8	8.7	-12.0	32.5	5.4	2.7	-15.9	34.4	-19.1	36.0	-22.1	37.5	-26.1	39.5		
			-12.3	22.0	4.9	1.5	-15.9	23.1	-18.7	24.0	-21.5	24.9	-25.1	26.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	2.3	2.0	5.1	8.1	12.7	15.1	18.2	17.5	13.8	9.5	5.5	3.6
	DBStd	6.76	5.03	5.00	4.05	3.81	3.74	3.31	3.28	3.07	2.35	3.34	3.64	3.82
	HDD10.0	1107	239	226	156	82	16	1	0	0	2	50	137	200
	HDD18.3	3309	497	458	410	306	179	109	44	55	138	274	384	456
	CDD10.0	923	1	1	4	26	98	153	254	231	115	35	3	2
	CDD18.3	85	0	0	0	1	4	11	41	28	1	0	0	0
	CDH23.3	898	0	0	0	7	61	132	398	285	14	0	0	0
	CDH26.7	247	0	0	0	0	4	24	129	88	2	0	0	0
Wind	WSAvg	4.0	5.2	4.7	4.8	3.9	3.5	3.3	3.4	3.4	3.5	3.9	4.2	4.7
Precipitation	PrecAvg	779	69	56	54	42	58	70	82	73	70	63	67	75
	PrecMax	1087	136	109	127	106	112	143	186	153	176	216	134	154
	PrecMin	592	7	15	9	0	22	18	30	21	11	12	3	18
	PrecStd	119	35	27	30	22	27	33	36	33	39	37	33	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	14.8	17.9	24.4	27.0	29.2	32.9	32.8	26.0	21.0	14.2	13.1
		MCWB	9.8	11.0	11.2	15.4	16.5	19.0	20.3	19.7	18.7	15.6	11.6	10.6
	2%	DB	10.8	11.9	14.8	20.2	25.1	26.9	30.1	29.0	22.1	18.1	12.8	11.9
		MCWB	8.4	8.7	9.9	12.9	15.6	18.0	20.0	18.6	16.9	14.0	10.9	9.9
	5%	DB	9.2	9.8	12.2	17.2	23.0	24.6	27.9	26.3	20.1	16.1	11.8	10.2
		MCWB	7.5	7.5	8.7	11.7	14.7	17.0	19.3	18.0	15.6	13.0	10.0	8.5
	10%	DB	8.0	8.1	10.8	14.8	20.1	22.1	25.1	24.0	18.2	14.8	10.2	9.1
		MCWB	6.5	6.3	8.1	10.5	13.7	16.1	18.1	17.3	14.3	12.1	8.8	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	11.0	11.5	15.5	17.8	21.6	21.6	21.1	19.7	16.1	12.3	11.2
		MCDB	11.8	14.1	15.9	23.1	24.5	26.8	29.2	28.6	23.8	19.2	13.5	12.6
	2%	WB	8.9	9.2	10.3	13.9	16.3	19.0	20.8	19.6	17.5	14.5	11.2	10.0
		MCDB	10.2	11.6	13.5	19.4	23.3	24.3	28.2	26.7	20.8	17.4	12.6	11.5
	5%	WB	7.6	7.5	9.4	12.3	15.2	17.5	19.7	18.7	16.1	13.3	10.2	8.8
		MCDB	9.0	9.3	11.9	16.6	21.6	23.2	26.0	24.7	19.3	15.8	11.4	10.1
	10%	WB	6.4	6.3	8.2	10.7	14.1	16.2	18.6	17.8	15.1	12.4	9.2	7.7
		MCDB	7.7	7.8	10.4	14.3	19.6	21.4	24.1	23.0	17.8	14.6	10.3	9.0
Mean Daily Temperature Range	5% DB	MDBR	4.9	6.1	7.2	9.0	10.3	9.4	10.3	10.0	8.1	7.2	5.6	4.5
		MCDBR	5.7	7.6	10.3	12.9	14.3	13.9	14.5	14.5	11.3	9.4	6.7	5.6
		MCWBR	5.0	5.7	6.6	7.2	7.2	7.0	6.4	6.6	6.7	6.3	5.4	4.9
	5% WB	MCDBR	5.4	7.0	9.3	11.7	13.0	12.5	12.9	12.3	9.2	8.5	5.6	5.3
		MCWBR	5.1	5.6	6.4	6.8	7.2	6.8	6.3	6.4	6.1	6.0	5.0	5.0
Clear-Sky Solar Irradiance	taub	0.323	0.343	0.374	0.407	0.406	0.405	0.419	0.411	0.385	0.358	0.338	0.316	
	taud	2.337	2.333	2.264	2.219	2.256	2.281	2.255	2.292	2.356	2.405	2.363	2.327	
	Ebn at Noon	652	756	804	822	843	847	826	810	787	730	627	585	
	Edh at Noon	61	82	106	126	127	126	127	116	97	75	59	52	
All-Sky Solar Radiation	RadAvg	0.61	1.29	2.42	3.98	4.91	5.20	4.97	4.15	3.01	1.71	0.76	0.48	
	RadStd	0.06	0.19	0.27	0.46	0.50	0.40	0.64	0.32	0.35	0.17	0.08	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-127	N/A	N/A

Nomenclature: See separate page